# Chapter 3 Study Guide: Angular Momentum

This guide summarizes the key concepts from Chapter 3, focusing on the principles needed to solve the selected exercises.

### 1. Key Definitions

- **Angular Momentum ($\sigma$):** The rotational equivalent of linear momentum. It measures the "quantity of rotational motion" of an object or system.
- **Formula for a Particle/Solid:**

$$\sigma = I \cdot \theta'$$

  - $\sigma$: Angular Momentum
  - $I$: Moment of Inertia (the rotational equivalent of mass, $\mathrm{kg} \cdot \mathrm{m}^2$)
  - $\theta'$: Angular Velocity ($\mathrm{rad/s}$)
- **Formula for a System:** The total angular momentum is the algebraic sum of the individual angular momenta.

$$\sigma_{\text{system}} = \Sigma\sigma_i = \sigma_1 + \sigma_2 + \dots$$

  - If all parts move together with the same $\theta'$, then $\sigma_{\text{system}} = (I_1 + I_2 + \dots)\theta' = I_{\text{total}} \cdot \theta'$
- **Units:** The S.I. unit for $\sigma$ is $\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}$.
- **Direction:** $\sigma$ is an algebraic quantity. Its sign (positive or negative) depends on the chosen positive direction of rotation. (See Ex 5 & 6)

### 2. The Two Core Principles

Your approach to *every* problem should start by asking: **"Is there a net external torque?"** The answer determines which of these two principles to apply.

#### Principle 1: Theorem of Angular Momentum (If $\Sigma M_{\text{ext}} \neq 0$)

This is the rotational version of Newton's Second Law. It states that a net external torque *changes* the angular momentum.

- **Formula:** $\Sigma M_{\text{ext}} = \frac{d\sigma}{dt}$
- **Analogy:** Just like $\Sigma \vec{F}_{\text{ext}} = \frac{d\vec{p}}{dt}$ (a force changes linear momentum).
- **When to use it:** When an *external* force or brake is applied for a period of time, causing the object to speed up or slow down.
- **Problem Type (e.g., Ex 12):**
  - A constant tangential force ($F$) creates a constant torque ($M = F \times r$).
  - Since $I$ is constant, $\Sigma M_{\text{ext}} = \frac{d(I\theta')}{dt} = I\frac{d\theta'}{dt} = I\theta''$.
  - You use this to find the angular acceleration ($\theta''$).
  - **Formula:** $\Sigma M_{\text{ext}} = I\theta''$

#### Principle 2: Conservation of Angular Momentum (If $\Sigma M_{\text{ext}} = 0$)

This is the most important concept in this chapter. If there is **no net external torque** on a system, its total angular momentum **cannot change**.

- **Condition:** $\Sigma M_{\text{ext}} = 0$
- **Result:** $\frac{d\sigma}{dt} = 0 \implies \sigma = \text{constant}$

- **Analogy:** Just like $\Sigma\vec{F}_{\text{ext}} = \vec{0} \implies \vec{p} = \text{constant}$.
- **Master Formula:** $\sigma_{\text{initial}} = \sigma_{\text{final}}$ or $I_i\theta'_i = I_f\theta'_f$
- **When to use it:** In all "before-and-after" scenarios where the system changes *internally* (pulling arms in, adding a mass, objects colliding, walking on a disk).

### 3. Common Scenarios for Conservation ($\sigma_i = \sigma_f$)

These are the most common applications you'll see in problems:

**Scenario 1: Internal Change in $I$** (Ex 7, 10, 14)

- **What happens:** The system's mass distribution changes *internally*.
- **Examples:**
  - A skater pulls her arms in (Ex 7).
  - A block slides from the center of a rod to the end (Ex 10).
  - A particle on a string is pulled into a smaller circle (Ex 14).
- **Analysis:**
  - $I_1\theta'_1 = I_2\theta'_2$
  - If $I$ decreases (arms in, $I_2 < I_1$), then $\theta'$ must increase ($\theta'_2 > \theta'_1$) to keep $\sigma$ constant.

**Scenario 2: Rotational "Inelastic Collision"** (Ex 8, 9)

- **What happens:** Two separate objects stick together and rotate with one common final velocity.
- **Examples:**
  - A particle is dropped onto a rotating disk (Ex 8).
  - Two flywheels clutch together (Ex 9).
- **Analysis:**
  - $\sigma_{\text{initial}} = \sigma_1 + \sigma_2$
  - $\sigma_{\text{final}} = \sigma_{\text{combined}} = (I_1 + I_2)\theta'_{\text{final}}$
  - **Equation:** $I_1\theta'_1 + I_2\theta'_2 = (I_1 + I_2)\theta'_f$
  - (Remember $\theta'_2$ might be negative if rotating in the opposite direction, as in Ex 9).

**Scenario 3: Starting from Rest** (Ex 16, 2.D)

- **What happens:** The system starts at rest, and then internal forces cause parts to move.
- **Example:** A man at rest on a disk starts to walk (Ex 16).
- **Analysis:**
  - $\sigma_{\text{initial}} = 0$ (since $\theta' = 0$).
  - Therefore, $\sigma_{\text{final}}$ must also be $0$.
  - $\sigma_{\text{final}} = \sigma_{\text{man}} + \sigma_{\text{disk}} = 0$
  - **Equation:** $\sigma_{\text{disk}} = -\sigma_{\text{man}}$
  - This means $I_{\text{disk}}\theta'_{\text{disk}} = -I_{\text{man}}\theta'_{\text{man}}$. Their angular momenta are equal and opposite, so they rotate in opposite directions.

### 4. Linear vs. Angular Analogy (The Key to Understanding)

| Concept | Linear (Ch 2) | Angular (Ch 3) |
|---|---|---|
| Inertia | Mass ($m$) | Moment of Inertia ($I$) |
| Velocity | $v$ | $\theta'$ |

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|---|---|---|
| Momentum | $p = m \cdot v$ | $\sigma = I \cdot \theta'$ |
| Cause of Change | Force ($F$) | Moment/Torque ($M$) |
| Newton's 2nd Law | $F_{\text{net}} = \frac{dp}{dt}$ | $M_{\text{net}} = \frac{d\sigma}{dt}$ |
| Conservation | If $F_{\text{net}} = 0$, $p$ is constant | If $M_{\text{net}} = 0$, $\sigma$ is constant |